

Sleep Deprivation and Memory Consolidation in Adults

Abstract

We examined how one night of total sleep deprivation affects next-day recall in a randomised crossover study of 40 healthy adults aged 18 to 35 years.

Introduction

Sleep is thought to support the consolidation of declarative memories, but the size of the effect in adults remains debated across laboratory studies.

Methods

Participants learned 60 word pairs in the evening and were tested the next morning after either a normal night of sleep or a night of supervised waking. Recall was scored blind to condition and analysed with a mixed effects model.

Results

Recall was 20% lower after a night without sleep than after normal sleep.

The effect was larger for weakly encoded pairs than for strongly encoded ones, suggesting that sleep preferentially stabilises fragile memory traces.

Discussion

These findings support an active role for sleep in memory consolidation.

References

Smith J, Lee K. Sleep and memory: a review. *Journal of Sleep Research*. 2019.

Garcia M. Total sleep deprivation and recall. *Neuropsychologia*. 2021.

Chen W, Patel R. Encoding strength and consolidation. *Memory*. 2020.